

Marcus N. Gomes Jr.

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Last update: August 31, 2026

Education

Ph.D., Hydraulic Engineering and Sanitation (Hydrology & Water Resources) <i>University of Sao Paulo, Sao Carlos School of Engineering</i> Dissertation: Advances in open source hydroinformatics for flood modeling and disaster education . GPA 4.0/4.0. Awarded best 2023 Ph.D. thesis in engineering at USP Sao Carlos School of Engineering.	Jan. 2020–Nov. 2023
Ph.D., Civil and Environmental Engineering[†] <i>University of Texas at San Antonio</i> Dissertation: Physics-based numerical models applied to sustainable urban design and real-time control for flood mitigation.	Jan. 2020–present
M.Sc., Hydraulic Engineering and Sanitation (Hydrology & Water Resources) <i>University of Sao Paulo, Sao Carlos School of Engineering</i>	Feb. 2018–Dec. 2019
B.Sc., Civil Engineering <i>State University of Maringa, Brazil</i>	Feb. 2013–Nov. 2017

Appointments

Postdoctoral Researcher <i>Stanford University, Department of Earth System Science</i> Stanford Doerr School of Sustainability Reconstructing recent floods in India and California using remote sensing and physically based hydrologic-hydrodynamic modeling; developing machine-learning inference methods to connect flood exposure with medical records. Advisor: Steven Gorelick, Ph.D.	Feb. 2026–present
Postdoctoral Research Associate <i>University of Arizona, Department of Hydrology and Atmospheric Sciences</i> Developed and applied hillslope-storage Boussinesq and land-surface modeling approaches for catchment hydrologic signatures, snow-influenced hydrology, and state-dependent effective material properties using high-resolution Landscape Evolution Observatory data. Advisor: Peter A. Troch, Ph.D.	Jan. 2024–Jan. 2026
Graduate Research Assistant <i>University of Texas at San Antonio, Civil and Environmental Engineering</i> Developed numerical models, stormwater control methods, and field monitoring programs for permeable pavement and urban water-quality projects funded by the City of San Antonio.	Jan. 2023–Dec. 2023
Graduate Research Assistant <i>University of Sao Paulo, Sao Carlos School of Engineering</i> Developed hydrologic/hydrodynamic models and supported field design and monitoring for bioretention and low-impact development systems funded by FAPESP.	Feb. 2018–Dec. 2019
Undergraduate Research Assistant <i>State University of Maringa, Civil Engineering</i> Brazil Conducted undergraduate research on flood mitigation in downtown Umuarama and proposed nature-based solutions to reduce urban inundation impacts.	June 2016–June 2017

Honors, Fellowships, and Research Metrics

Award	Best Ph.D. thesis of 2023 in engineering at the University of Sao Paulo, Sao Carlos School of Engineering, including a R\$ 10,000 award.
Google Scholar	709 citations (since 2021: 702); h-index 14; i10-index 16 (as of July 19, 2026). Profile .
Moore Foundation	Offered Gordon and Betty Moore Foundation scholarship / fellowship support for 2025–2026 (approximately \$70,000); not accepted.
City of San Antonio	Ph.D. project support, approximately \$153,000 USD. Project: Demonstrating the Environmental Benefits of Permeable Paved Surfaces over the Edwards Aquifer .

[†]Dissertation defense on hold while on current postdoctoral appointment due to visa requirements.

CAPES	Ph.D. scholarship (Grant 88887.663281 / 2022-00), R\$ 74,400. Funding source.
FAPESP	M.Sc. scholarship (Process 18/20865-0), R\$ 58,116. Funding source.
FAPESP Regular Research Grant	Project 2017/21940-0, 2018–2020, R\$ 158.243,56.
Undergraduate support	Free tuition by state university scholarship, approximately R\$ 105,000.00.

Peer-reviewed journal publications

* indicates mentee first author under my mentorship.

2026

36. **Gomes Jr., M. N.**, Elsherif, S. M., Benites, I. M. *, Taha, A. F., and Giacomoni, M. H. (2026). *Modeling and Design Optimization of Looped Water Distribution Networks using MS Excel: Developing the Open-Source X-WHAT Model*. **Water Resources Management**, 40, 514. DOI: [10.1007/s11269-026-04817-7](#).
35. **Gomes Jr., M. N.** and Castro, M. A. R. A. (2026). *A Continental-scale XGBoost Framework for Multi-Hazard Susceptibility and Socioeconomic Exposure in South America*. **Sustainable Cities and Society** (Accepted).
34. Sánchez, M. H. *, **Gomes Jr., M. N.**, Castillo, L. M. R., Castro, M. A. R. A., and Mendiondo, E. M. (2026). *Assessment of urban vulnerability to floods under current and future demographic and climate scenarios*. **International Journal of Disaster Risk Reduction**, 106337. DOI: [10.1016/j.ijdr.2026.106337](#).
33. **Gomes Jr., M. N.** (2026). *LID-Design Atlas: Mapping Hydrologic Performance and Cost Trade-Offs in Infiltration-Based LID Design*. **Journal of Hydrologic Engineering**, 31(5). DOI: [10.1061/JHYEFF.HEENG-6901](#).
32. Navarro, F. A. R., **Gomes Jr., M. N.**, & Mendiondo, E. M. (2026). *Risk management framework for urban catchments under non-stationary analysis: Mitigation and adaptation scenarios in a Brazilian case study*. **Urban Water Journal**. DOI: [10.1080/1573062X.2026.2682572](#).
31. **Gomes Jr., M.N.**, 2026. Accounting for spatial runoff variability in LID design for urban catchments: model and software development. **Brazilian Journal of Water Resources**, 31, p.e10. DOI: [10.1590/2318-0331.312620250060](#)
30. **Gomes Jr., M. N.**, Bauser, H. H., Troch, P. A. *Identifying the State Dependence of Effective Material Properties in a Simplified Hydrologic Hillslope Model*. **Water Resources Research**, 2026. DOI: [10.1029/2025WR041468](#).
29. Brandão, A.R., Castro, M.A., Sánchez, M.H., **Gomes Jr., M.N.**, Uchôa, J.G.S., Vaz, I.C., Ghisi, E., Anache, J.A., Wendland, E.C., Oliveira, P.T. and Mendiondo, E.M., 2026. Flood Event Escalation and Urban Drainage Design Implications Under Nonstationary Rainfall in São Paulo State, Brazil. **Water**, 18(5), p.561. DOI: [10.3390/w18050561](#)
28. Castro, M., **Gomes Jr., M. N.**, Montenegro, S. (2026) *Assessing the Socioeconomic Benefits of the Agreste Water Supply System in Pernambuco, Brazil*. **Brazilian Journal of Water Resources**. DOI: [10.1590/2318-0331.312620250154](#).
27. Brasil, J., Papagiannakis, A. T., Johnson, D. W., **Gomes Jr., M. N.**, and Giacomoni, M. H. (2026). *Field Scale Test Bed for Assessment of the Surface Runoff and Water Quality Performance of Permeable Pavements*. **Journal of Hydrologic Engineering**, 31(4), p.04026014. DOI: [10.1061/JHYEFF.HEENG-6602](#).
26. Sousa, M. R. *, **Gomes Jr., M. N.**, Mendiondo, E. M. *Hydrological–Hydrodynamic Modeling of Climate-Induced Urban Flooding of Design Storms Using HydroPol2D: A Case Study in São Carlos, Brazil*. **Brazilian Journal of Water Resources**. DOI: [10.1590/2318-0331.312620250090](#).

2025

24. **Gomes Jr., M.N.**, Castro, M.A., Castillo, L.M., Sánchez, M.H., Giacomoni, M.H., de Paiva, R.C. and Bates, P.D., 2025. *Spatio-temporal performance of 2D local inertial hydrodynamic models for urban drainage and dam-break applications*. **Journal of Hydrology**, p.134661. DOI: [10.1016/j.jhydrol.2025.134661](#)
23. Rápalo, L. M. C., **Gomes Jr., M. N.**, Mendiondo, E. M. *Multiple Degrees of Human Instability Due to Urban Overland Flow Within the 21st Century: An Urban Watershed Case Study in Brazil*. **International Journal of Disaster Risk Reduction**. DOI: [10.1016/j.ijdr.2025.105931](#).
22. **Gomes Jr., M. N.**, Brasil, A., Johnson, D. W., Papagiannakis, A. T., Giacomoni, M. H. (2025) *A Coupled Darcy–Richards Framework for Hydrological Modeling of Permeable Pavements, Green Roofs, and Bioretention Systems*. **Environ-**

mental Modelling and Software.
DOI: [10.1016/j.envsoft.2025.106766](https://doi.org/10.1016/j.envsoft.2025.106766).

21. **Gomes, M.N.**, Jalihal, V., Castro, M. and Mendiondo, E.M., 2025. *Exploring the impact of rainfall temporal distribution and critical durations on flood hazard modeling*. **Natural Hazards**, 121(9), pp.10989–11012. [10.1007/s11069-025-07186-3](https://doi.org/10.1007/s11069-025-07186-3).
20. **Gomes Jr, M.N.**, do Lago, C.A. and Mendiondo, E.M., 2025. *A Simple Method for Designing Infiltration Low Impact Development Techniques Considering Effects of Urbanization and Climate Change*, **Brazilian Journal of Water Resources (RBHR)** (2025).
DOI: [10.1590/2318-0331.302520240106](https://doi.org/10.1590/2318-0331.302520240106)
19. Vasconcelos, J.G., **Gomes Jr, M.N.**, Oliveira, P.T.S., Yang, D. and Fang, X., 2025. *Reformulating the Rational Method Considering Dissimilar Land Use Types*. **Journal of Irrigation and Drainage Engineering**, 151(4), p.04025017. [10.1061/JIDEDH.IRENG-10431](https://doi.org/10.1061/JIDEDH.IRENG-10431).

2024

18. **Gomes Jr**, Marcus Nóbrega, Ahmad F. Taha, Luis Miguel Castillo Rápalo, Eduardo Mario Mendiondo, and Marcio Hofheinz Giacomoni. “Real-time regulation of detention ponds via feedback control: Balancing flood mitigation and water quality.” **Journal of Hydrology** 643 (2024): 131866. DOI: [10.1016/j.jhydrol.2024.131866](https://doi.org/10.1016/j.jhydrol.2024.131866).
17. **Gomes Jr**, Marcus Nóbrega, Maria de Andrade Rocha Alencar Castro, Pedro Gustavo Câmara da Silva, Marcio Hofheinz Giacomoni, and Eduardo Mario Mendiondo. *Increasing flood awareness through dam-break serious games*. **International Journal of Disaster Risk Reduction** 108 (2024): 104543. DOI: [10.1016/j.ijdrr.2024.104543](https://doi.org/10.1016/j.ijdrr.2024.104543).
16. Rápalo, Luis MC, Marcus N. **Gomes Jr**, and Eduardo M. Mendiondo. “Developing an open-source flood forecasting system adapted to data-scarce regions: A digital twin coupled with hydrologic-hydrodynamic simulations.” **Journal of Hydrology** (2024): 131929. DOI: [10.1016/j.jhydrol.2024.131929](https://doi.org/10.1016/j.jhydrol.2024.131929).
15. Ambrogi Ferreira Do Lago, Cesar, Jose Artur Teixeira Brasil, Marcus Nóbrega **Gomes**, Eduardo Mario Mendiondo, and Marcio H. Giacomoni. “Improving pluviial flood mapping resolution of large coarse models with deep learning.” **Hydrological Sciences Journal** 69, no. 5 (2024): 607–621. DOI: [10.1080/02626667.2024.2329268](https://doi.org/10.1080/02626667.2024.2329268).

2023

14. M. N. **Gomes Jr.**, C. A. F. do Lago, L. M. C. Rápalo, P. T. S. Oliveira, M. H. Giacomoni, and E. M. Mendiondo, “HydroPol2D – Distributed Hydrodynamic and Water Quality Model: Challenges and Opportunities in Poorly-Gauged Catchments,” **Journal of Hydrology**, 2023. DOI: [10.1016/j.jhydrol.2023.129982](https://doi.org/10.1016/j.jhydrol.2023.129982).
13. M. N. **Gomes Jr.**, L. M. C. Rapalo, P. T. S. Oliveira, M. H. Giacomoni, C. A. F. do Lago, and E. M. Mendiondo, “Modeling unsteady and steady 1-d hydrodynamics under different hydraulic conceptualizations: Model/software development and case studies,” **Environmental Modeling and Software**, 2023. DOI: [10.1016/j.envsoft.2023.105733](https://doi.org/10.1016/j.envsoft.2023.105733).
12. M. N. **Gomes Jr.**, M. H. Giacomoni, M. B. de Macedo, C. A. F. do Lago, J. A. Teixeira, T. R. Pereira, and E. M. Mendiondo, “A modeling framework for bioretention analysis: Assessing the hydrologic performance under system’s uncertainty,” **Journal of Hydrologic Engineering**, DOI: [10.1061/JHYEFF.HEENG-5705](https://doi.org/10.1061/JHYEFF.HEENG-5705) 2023.
11. A. S. Ballarin, J. G. S. M. Uchôa, M. S. dos Santos, A. Almagro, I. P. Miranda, P. G. C. da Silva, M. N. **Gomes Jr.**, E. Wendland, P. T. S. Oliveira, “Brazilian water security threatened by climate change and human behavior”. **Water Resources Research**, DOI: [10.1029/2023WR034914](https://doi.org/10.1029/2023WR034914) 2023.
10. do Lago, C.A., Giacomoni, M.H., Bentivoglio, R., Taormina, R., **Junior**, M.N.G. and Mendiondo, E.M., 2023. *Generalizing rapid flood predictions to unseen urban catchments with conditional generative adversarial networks*. **Journal of Hydrology**, 618, p.129276, DOI: [10.1016/j.jhydrol.2023.129276](https://doi.org/10.1016/j.jhydrol.2023.129276).

2022

9. M. N. **Gomes Jr.**, M. H. Giacomoni, A. F. Taha, and E. M. Mendiondo, “Flood risk mitigation and valve control in stormwater systems: State-space modeling, control algorithms, and case studies,” **Journal of Water Resources Planning and Management**, vol. 148, no. 12, p. 04022067, 2022. [Online]. Available: [10.1061/\(ASCE\)WR.1943-5452.0001588](https://doi.org/10.1061/(ASCE)WR.1943-5452.0001588)
8. M. Batalini de Macedo, M. Nobrega **Gomes Júnior**, T. R. Pereira de Oliveira, M. H. Giacomoni, M. Imani, K. Zhang, C. Ambrogi Ferreira do Lago, and E. M. Mendiondo, “Low impact development practices in the context of united nations sustainable development goals: A new concept, lessons learned and challenges,” **Critical Reviews in Environmental Science and Technology**, vol. 52, no. 14, pp. 2538–2581, 2022, DOI: [10.1080/10643389.2021.1886889](https://doi.org/10.1080/10643389.2021.1886889)

7. J. A. T. Brasil, M. B. de Macedo, T. R. P. de Oliveira, F. G. Ghiglieno, V. C. B. de Souza, G. Marinho e Silva, M. N. **Gomes Júnior**, F. A. A. de Souza, and E. M. Mendiondo, "Can we scale digital twins of nature-based solutions for stormwater and transboundary water security projects?" **Journal of Hydroinformatics**, 2022, DOI: [10.2166/hydro.2022.142](https://doi.org/10.2166/hydro.2022.142)
6. M. Batalini de Macedo, M. N. **Gomes Júnior**, V. Jochelavicius, T. R. P. de Oliveira, and E. M. Mendiondo, "Modular design of bioretention systems for sustainable stormwater management under drivers of urbanization and climate change," **Sustainability**, vol. 14, no. 11, p. 6799, 2022, [10.3390/su14116799](https://doi.org/10.3390/su14116799).
5. Jochelavicius, V., E. M. Mendiondo, M. N. **Gomes Jr.**, and M. B. de Macedo, "Construction of Intensity-Duration-Frequency curves with future climate change scenarios for the city of São Carlos - SP aiming at the design of compensatory techniques," **Revista DAE**, vol. 147, p. 19-32, [10.36659/dae.2022.065](https://doi.org/10.36659/dae.2022.065) (In Portuguese).

2021

4. J. Brasil, M. Macedo, C. Lago, T. Oliveira, M. **Júnior**, T. Oliveira, and E. Mendiondo, "Nature-based solutions and real-time control: Challenges and opportunities," **Water**, vol. 13, no. 5, p. 651, 2021, DOI: [10.3390/w13050651](https://doi.org/10.3390/w13050651).
3. T. R. P. de Oliveira, M. B. de Macedo, T. H. Oliveira, C. A. F. do Lago, M. N. **Gomes Jr.**, J. A. T. Brasil, and E. M. Mendiondo, "Different configurations of a bioretention system focused on stormwater harvesting in Brazil," **Journal of Environmental Engineering**, vol. 147, no. 12, p. 04021058, 2021, [https://doi.org/10.1061/\(ASCE\)EE.1943-7870.0001938](https://doi.org/10.1061/(ASCE)EE.1943-7870.0001938)

2020

2. M. N. **Gomes Jr.**, P. H. B. Alves, E. M. Mendiondo, and L. F. R. Reis, "Statistical, visual and non-parametric analyses for the optimization of IDF curve fitting and construction of hydraulic works design abacuses: case study in São Carlos - SP". **Revista DAE**, DOI: [10.36659/dae.2021.013](https://doi.org/10.36659/dae.2021.013) 2020 (In Portuguese).
1. K. McClymont, D. G. F. Cunha, C. Maidment, B. Ashagre, A. F. Vasconcelos, M. B. de Macedo, M. F. N. Dos Santos, M. N. G. **Júnior**, E. M. Mendiondo, A. P. Barbassa et al., "Towards urban resilience through sustainable drainage systems: A multi-objective optimisation problem", **Journal of Environmental Management**, vol. 275, p. 111173, 2020, DOI: [10.1016/j.jenvman.2020.111173](https://doi.org/10.1016/j.jenvman.2020.111173)

Publications under review or in preparation

First-author manuscripts

10. **Gomes Jr., M. N.**, Ballarin, A. S., and Oliveira, P. T. S. Brazilian Flood-Mitigation Infrastructure Is Underdesigned Before Climate Change: Required Peaks More Than Double and Storage More Than Quadruples Under the Late-Century SSP5-8.5 Ensemble Mean. **Earth's Future** (In preparation).
9. **Gomes Jr., M. N.**, Castro, M. A. R. A., Nguyen, A., Wang, A., Gorelick, S. M. Global Inequalities in Urban Hazard Risk. **Nature Communications** (In preparation).
8. **Gomes Jr., M. N.**, Castro, M., Ballarin, A., Troch, P., Almagro, A., and Oliveira, P. T. S. Levels and Indicators for Monitoring Inundation Across Rivers (LIMIAR): Hydraulic Flood Reconstruction in Brazil. **Hydrology and Earth System Sciences** (Under preparation).
7. **Gomes Jr., M. N.**, Ballarin, A. S., Oliveira, P. T. S. Time Distribution of Heavy Rainfall in Brazil: Empirical Huff Curves from 290,164 Sub-Daily Storm Events. **Water Resources Research** (Under Review).
6. **Gomes Jr., M. N.** A Two Gamma Reduced Analytical Hydrograph for SCS-CN Huff Design Storms. **Journal of Hydrologic Engineering** (Under Review).
5. **Gomes Jr., M. N.** A Nationally Consistent Assessment of Intensity–Duration–Frequency Curves for Brazil Using Long-Term Gridded and Satellite Rainfall Data. **Physics and Chemistry of the Earth** (Under Review).

Coauthored manuscripts

4. Bauser, H. H., Kim, M., Troch, P. A., and LEO Hydrology Team (2013–2026; including **Gomes Jr., M. N.**). Addressing the Hydrologic Scaling Challenge Using Experimental Hillslopes: 13 Years of Discovery at the Landscape Evolution Observatory (LEO) at Biosphere 2. **Water Resources Research** (Under review).
3. Castro, M. A. R. A., **Gomes Jr., M. N.**, Mendiondo, E. M. Probabilistic Dam Break Flood Mapping via Monte-Carlo Simulation. **Journal of Hydrology**. DOI: <https://essopenarchive.org/doi/full/10.22541/essoar.175181464.43589172> (Submitted).

2. Castro, M. A. R. A., **Gomes Jr., M. N.**, Mendiondo, E. M. *Trade-Offs Between Full Momentum, Diffusive Wave, and Local-Inertial Solvers for Dam-Break Flood Modeling*. **Environmental Modelling & Assessment** (Under review).
1. Brasil, J. T., **Gomes Jr., M. N.**, Johnson, D. W., Papagiannakis, A. T., Mendiondo, E. M., Giacomoni, M. H. *A Digital Twin Framework for Real-Time Probabilistic Forecasting of Permeable Pavement Performance*. **Journal of Hydrology** (Under Review).

Selected Research Products (n = 7 of 24 public repositories on GitHub)

Product / repository	Scope and public link
HydroPol2D	Distributed 2D hydrologic-hydrodynamic and water-quality model for overland flow, infiltration, groundwater recharge, shallow groundwater, reservoirs, dam-breaks, snow, and pollutant transport. GitHub: marcusnobrega-eng/HydroPol2D ; docs: HydroPol2D-docs ; article: <i>Journal of Hydrology</i> , DOI: 10.1016/j.jhydrol.2023.129982 .
MIRA	Interactive South America multi-hazard platform for flood, drought, wildfire, socioeconomic vulnerability, and city-priority analysis at 1km resolution. Viewer: riskframeworksa.github.io/data_viewer .
GRIDF	Gridded intensity-duration-frequency curves for Brazil, including national methods for IDF estimation and visualization. GitHub: marcusnobrega-eng/GRIDF ; Earth Engine app: gridf-br .
Huff Curves BR Atlas	A web-based tool applied to all rainfall gauges in Brazil to visualize how rainfall dynamics change over time. For a same rainfall volume and duration, the temporal distribution can increase peak flows in approximately 10%, for example. Viewer: Huff_Curves_BR/docs .
LIMIAR	A web-based river-flood reconstruction tool for all monitored stream gauges in Brazil by using data such as (i) river cross-sections, (ii) water level, (iii) discharge, and (iv) rating curves. GitHub: marcusnobrega-eng/LIMIAR ; docs: marcusnobrega-eng.github.io/LIMIAR .
DRAIN-LID	1D finite-volume model for simulating unsaturated flow in porous media based on Darcy-Richards equations. Specialized hydrodynamic tool for permeable pavements, green roofs, bioretention systems. GitHub: marcusnobrega-eng/DRAIN-LID ; docs: marcusnobrega-eng.github.io/DRAIN-LID .
India Flood Inventory Toolbox (IFI)	By using advanced LLMs scripting, the IFI Toolbox Website reconstruct all recorded flood events in India, categorizing attributes such as damages, fatalities, people affected, and etc from 1970 to 2023. Toolbox for analysis of India Flood Inventory records and related flood-database tools. GitHub: marcusnobrega-eng/IFI_Toolbox ; dashboard/docs: marcusnobrega-eng.github.io/IFI_Toolbox .

Conference proceedings

* indicates student mentee.

Full articles in conference proceedings

27. Brasil, J.A., Papagiannakis, A.T., Giacomoni, M.H., Johnson, D., **Gomes Jr, M.N.** and Mendiondo, E.M., 2023. Hydrologic and Water Quality Treatment Performance of Four Permeable Pavement Surfaces. In *International Low Impact Development Conference 2023* (pp. 114-122). *Award of Best Paper*.
26. M. N. **Gomes Jr.**, Giacomoni, M.H., Taha, A.F. and Mendiondo, E.M., 2022, August. *Model Predictive Control for Stormwater Reservoirs: Investigating Effects of Climate Change and Urbanization*. **IEEE Conference on Control Technology and Applications (CCTA)** (pp. 691-698). IEEE.
25. M. N. **Gomes Jr.**, P. A. Braga, M. B. de Macedo, E. M. Mendiondo, "Preliminary design of detention ponds using specific design discharge and orifice stage discharge relationship for different climate patterns". **Brazilian Symposium of Water Resources**, 2021.
24. M. N. **Gomes Jr.**, P. A. Braga, E. M. Mendiondo, "Impact of rainfall changes on the Pacaembu-SP detention reservoir: assessment of water security regarding flood risk". **Brazilian Symposium of Water Resources**, 2019 (In Portuguese).
23. M. N. **Gomes Jr.**, E. M. Mendiondo, F. Dornelles, A. T. Papagiannakis, "Permeable Pavement Hydrological Model to Assess the Long-Term Efficiency of Maintenance Using High-Resolution Temperature and Rainfall Data". **World Environmental and Water Resources Congress**, 2021, 1103-1117.
22. M. N. **Gomes Jr.**, E. O. Pavan, L. M. C. Rápallo, M. H. Giacomoni, E. M. Mendiondo, "Bidirectional Hydrodynamic and Water Quality Model (2DCAWQ): Modeling Challenges in Basins with Scarce Data: Application in the Tijuco Preto

- São Carlos Basin". **XIV National Meeting on Urban Waters and IV Symposium on Revitalization of Urban Rivers**, 2022 (In Portuguese).

21. M. N. **Gomes Jr.**, M. H. Giacomoni, E. M. Mendiondo, "*The role of raster resolution into overland flow and total suspended solids modeling in small urban catchments*". **Brazilian Symposium of Water Resources**, (2021).
20. M. N. **Gomes Jr.**, M. H. Giacomoni, A. T. Papagiannakis, "*Spatial Assessment of Overland Flow, Pollutant Concentration, and First Flush Using a 2D Non-Point Source Pollution and Hydrological Model for Urban Catchments*". **World Environmental and Water Resources Congress**, 2021, 397-413.
19. M. N. **Gomes Jr.**, M. H. Giacomoni, E. M. Mendiondo, "*Bioretention Sizing via Process-Based Simulation: Generalized TC-Hydro Model Applied in Design Conditions*". **XIV National Meeting on Urban Waters and IV Symposium on Revitalization of Urban Rivers**, 2022 (In Portuguese).
18. M. N. **Gomes Jr.**, L. M. C. Rápalo, E. M. Mendiondo, "*A Serious Game for societal risk perception of dam-break flood assessment using a hydrodynamic model*", **Latin American Hydraulic Congress**, (2022). *Award of Best Paper*.
17. M. N. **Gomes Jr.**, C. A. F do Lago, M. H. Giacomoni, E. M. Mendiondo, "*Assessing the role of cross-section shape for flood routing under compound flood events using a hydrodynamic model*", **Latin American Hydraulic Congress**, (2022).
16. T. R. P. de Oliveira, M. N. **Gomes Jr.**, M. B. de Macedo, E. M. Mendiondo, "*Análise da sensibilidade do potencial matricial em protótipo de biorretenção*". **Brazilian Symposium of Water Resources**, 2019 (In Portuguese).
15. CASTRO, M.A.R.A.; **Gomes Jr.**, M.N.; SANCHEZ, M.H.; FILHO, P.B.S., MENDIONDO, E.M. "*Review of the Risk Index Applied to Dam Failure Simulation in Brazil*". **XXV Brazilian Symposium of Water Resources**. Anais. Aracaju (SE), 2023 (In Portuguese).
14. SANCHEZ, M.H.; **Gomes Jr.**, M.N.; CASTRO, M.A.R.A.; MENDIONDO, E.M. "*River flow forecasting methods: A review*". **XXV Brazilian Symposium of Water Resources**. Anais. Aracaju (SE), 2023.
13. Milena R. de Sousa*, M. N. **Gomes Jr.**, E. O. Pavan, L. M. C. Rápalo, F. A. R. Nararro, E. M. Mendiondo, "*Physically Distributed Modeling Based on Urban Watersheds: Hydropol2D Model Applied to São Carlos (SP)*", **XXV Brazilian Symposium of Water Resources**. Proceedings. Aracaju (SE), 2023 (In Portuguese).
12. Enrico O. Pavan*, M. N. **Gomes Jr.**, L. M. C. Rápalo, M. R. de Sousa, M. S. dos Santos F. A. R. Nararro, E. M. Mendiondo, "*Physically Based Modeling of Large Basins - Challenges and Opportunities for Applying The Hydropol2D Model*", **XXV Brazilian Symposium of Water Resources**. Proceedings. Aracaju (SE), 2023 (In Portuguese).
11. Lucas F. Roysen*, M. N. **Gomes Jr.**, E. O. Pavan, E. M. Mendiondo, "*RTC-Stormwater Model: Advances In Flood Modeling And Control Of Urban Drainage Devices*", **XXV Brazilian Symposium of Water Resources**. Proceedings. Aracaju (SE), 2023 (In Portuguese).
10. A. S. Ballarin, J. G. S. M. Uchoa, M. S. dos Santos, A. Almagro, I. P. Miranda, P. G. C. da Silva, G. J. da Silva, M. N. **Gomes Jr.**, E. C. Wendland, P. T. S. de Oliveira, "*The Future of Water Security In Brazil: Separating The Impacts Of Climate Change And Water Demand*", **XXV Brazilian Symposium of Water Resources**. Proceedings. Aracaju (SE), 2023 (In Portuguese).
9. M. S. dos Santos, M. N. **Gomes Jr.**, D. A. Bressianim L. L. Ladeira, M. R. Benso, E. M. Mendiondo "*Review of the Implementation of Model Predictive Control (MPC) In Urban Reservoirs*", **XXV Brazilian Symposium of Water Resources**. Proceedings. Aracaju (SE), 2023 (In Portuguese).
8. G. M. e Silva, M. R. Benso, P. G. C. da Silva, M. N. **Gomes Jr.**, E. M. Mendiondo, "*Serious Game on Water Risks For Planetary Health: A New Model For Knowledge Co-Production*", **XXV Brazilian Symposium of Water Resources**. Proceedings. Aracaju (SE), 2023 (In Portuguese).
7. L. M. C. Rápalo, M. N. **Gomes Jr.**, D. A. Bressiani, M. R. Benso, E. M. Mendiondo "*Threats of Urban Flash Floods on Human Stability for The Next Century*", **XXV Brazilian Symposium of Water Resources**. Proceedings. Aracaju (SE), 2023.
6. I. M. Benites, M. N. **Gomes Jr.**, A. Botari, J. C. Botari, L. Vanalli, "*Analysis of the Urban Drainage System: Case Study of the Stormwater Galleries at the intersection of Rua Governador Nei Braga and Avenida Brasil in the city of Umuarama-PR*". **XVII Safety, Health and Environment World Congress**, 2017 (In Portuguese).

Extended abstracts

5. M. N. **Gomes Jr.**, M. H. Giacomoni, E. M. Mendiondo, A. F. Taha, "*Real-Time Control of Stormwater Reservoirs for Flood Risk Mitigation*". **The 2nd International Symposium on Water System Operations**, 2021.

4. M. N. **Gomes Jr.**, T. R. de Oliveira, T. H. Oliveira, M. B. de Macedo, M. H. Giacomoni, E. M. MENDIONDO, "Nature-based solutions for sustainable stormwater: a model approach and sensitivity analysis for bioretention design using Green and Ampt and reservoir flood routing". **Second International Conference of Water, Megacities, and Global Change, UNESCO**, 2021.

Abstracts

5. **Gomes Jr., M. N.**, Bauser, H. H., and Troch, P. A., *Characterizing the state dependence of effective material properties in a simplified hydrologic hillslope model*, **AGU24 Annual Meeting, Washington, DC**, 2024.
4. **Gomes Jr. M. N.**, H. Bauser, P. A. Troch, *Identifying the state dependence of effective material properties in a simplified hydrologic hillslope model*, **The Conference on Computational Methods on Water Resources (CMWR), Tucson**, 2024.
3. M. N. **Gomes Jr.**, C. A. F. do Lago, M. H. Giacomoni, E. M. MENDIONDO, "Expanding the 2-Dimensional Green-Ampt and Non-linear Reservoir Hydrological Model from SWMM to MATLAB®". **AGU Fall Meeting, New Orleans**, 2021.
2. C. A. F. do Lago, M. N. **Gomes Jr.**, E. M. MENDIONDO, M. H. Giacomoni, "Application of Artificial Neural Networks to Predict Water Surface Elevation". **AGU Fall Meeting, 2021, New Orleans**.
1. F. A. A. Souza, E. M. MENDIONDO, D. Taffarello, A. C. S. Buarque, N. Bhattacharya-Mis, C. E. Restrepo-Estrada, M. C. Fava, A. C. B. Delbem, S. C. Andrade, N. Abe, F. G. Abreu, M. N. **Gomes Jr.**, and M. B. de Macedo. *How to Bridge the Gaps among DRR, SDG, UPH, MOXXI/CANDHY and Panta Rhei by Volunteered Information of Human-Centered Observatories?* **AGU Fall Meeting, San Francisco, CA**, 2019, abstract H11O-1725 (presented by E. M. MENDIONDO).

Talks and short courses

- **Research talk**, Stanford University, Stanford, CA, Feb. 20, 2026. "Designing Open-Source Tools for Simulating Hydrological Processes: Challenges and Lessons from Building from Scratch."
- **Invited research talk**, Landscape Evolution Observatory / Digital Twin Conference, University of Arizona, Tucson, AZ, Dec. 11, 2025. "Identifying Effective Soil Properties in the Landscape Evolution Observatory."
- **Invited lecture**, **University of Arizona Hydrology and Atmospheric Sciences master's program**, "Fluid Dynamics - Bernoulli's Lecture," Tucson, AZ, Sept. 12, 2025.
- **AGU24 Annual Meeting**, Washington, DC, Dec. 2024. Poster: "Characterizing the State Dependence of Effective Material Properties in a Simplified Hydrologic Hillslope Model."
- **HydroPol2D short course**, University of Sao Paulo, Sao Carlos, Brazil, 2024. Model setup, running, and calibration.
- **HydroPol2D short course for graduate students**, University of Texas at San Antonio, San Antonio, TX, Nov. 21, 2024.
- **Invited class presentation**, **Water Resources System Analysis**, University of Texas at San Antonio, "1D hydrodynamic simulations with HydroHP," San Antonio, TX, Oct. 8, 2024.
- **Conference on Computational Methods in Water Resources**, Tucson, AZ, Oct. 1, 2024. Oral and poster presentations: "Identifying the state dependence of effective material properties in a simplified hydrologic hillslope model" and "Large Scale Fully-Distributed Hydrologic-Hydrodynamic Modeling: Developing an Early-Warning Tool."
- **HydroPol2D tutorial for graduate students**, University of Sao Paulo, Sao Carlos, Brazil, Sept. 1, 2023. Three-day course on fully distributed hydrologic-hydrodynamic modeling.
- **XXV Brazilian Symposium of Water Resources**, Aracaju, Brazil, 2023. Coauthored and supported presentations by undergraduate mentees Enrico Pavan, Lucas F. Roysen, and Milena Rosa de Sousa on HydroPol2D, RTC-Stormwater, and urban flood modeling.
- **Invited talk**, **Hydraulic System Course**, University of Sao Paulo, Sao Carlos, Brazil, Apr. 1, 2023. Presentation on HydroHP-1D and HydroPol2D applications in Amazon basin and urban catchments.
- **Invited undergraduate class presentation**, Vanderbilt University, Nashville, TN, Nov. 29, 2022. Two classes on RTC-Stormwater and real-time control of catchments, reservoirs, and channels.
- **Latin American Hydraulic Congress**, Foz do Iguacu, Brazil, Nov. 2022. Presented two HydroHP-1D papers; one selected as best paper in its category.
- **National Urban Water Meeting and Symposium on Revitalization of Urban Rivers**, Brasilia, Brazil, Sept. 2022. Presented two conference papers.
- **Second International Conference Water, Megacities and Global Change**, UNESCO, Paris, France, Dec. 2021. Presentation on nature-based solutions and bioretention modeling.
- **Brazilian Symposium of Water Resources**, Belo Horizonte, Brazil, Nov. 2021. Presentation on raster resolution and water-quality modeling in a fully distributed hydrologic-hydrodynamic model.
- **2nd International Symposium on Water System Operations**, Bristol, UK, Sept. 2021. "Real-Time Control of Stormwater Reservoirs for Flood Risk Mitigation."
- **World Environmental and Water Resources Congress**, Las Vegas, NV / online, June 2021. Presented papers on permeable pavement hydrology and distributed runoff/pollutant modeling.

- **AGU Fall Meeting**, New Orleans, LA, Dec. 2021. “Expanding the 2-Dimensional Green-Ampt and Non-linear Reservoir Hydrological Model from SWMM to MATLAB.”
- **Advanced Excel short course**, University of Sao Paulo, Sao Carlos School of Engineering, Sao Carlos, Brazil, 2019. One-week course on advanced Excel methods for engineering calculations and data analysis.
- **Brazilian Symposium of Water Resources**, Foz do Iguacu, Brazil, Dec. 2019. Presented papers on inverse modeling of the Pacaembu detention reservoir and orifice discharge sensitivity.

Teaching Experience

CIRTL teaching-development course	2025
<i>University of Arizona</i> Completed the Center for the Integration of Research, Teaching, and Learning (CIRTL) program course through the University of Arizona.	
Teaching Assistant, Fundamentals of Surface Water Hydrology	2024
<i>University of Arizona, Department of Hydrology and Atmospheric Sciences</i> Reviewed and graded assignments, projects, and exams for a graduate hydrology course taught by Peter A. Troch.	
Teaching Assistant, Hydrology I and II for Civil Engineering	2022–2023
<i>University of Sao Paulo</i> Developed supporting materials, spreadsheets, and numerical models; taught selected applied hydrology classes for undergraduate students.	
Teaching Assistant, Hydrology I and II for Civil Engineering	2018–2019
<i>University of Sao Paulo</i> Developed supporting materials and computational exercises for applied hydrology courses.	
Instructor, Financial Mathematics	2015–2017
<i>CIEE (Center for School-Business Integration, Brazil)</i> Taught financial mathematics in the youth apprenticeship program.	
Instructor, Mathematics, Physics, and CAD-Assisted Technical Drawing	2015–2017
<i>Yesbras</i> Taught mathematics, physics, and technical drawing with CAD support.	
Teaching Assistant, Physics I and II	2014–2016
<i>State University of Maringa</i> Led problem-solving sessions and exam preparation for civil and environmental engineering students.	
Teaching Assistant, Physics, Mathematics, and Chemistry	2013–2017
<i>Curso e Colegio Alfa Preparatory School</i> Supported pre-college preparation in physics, mathematics, and chemistry.	

Funding and Proposal Experience

- **NSF Collaborative Research: Enabling Safe and Efficient Restoration of Drinking Water Systems through Multiphase Flow Modeling and Control** (2026–2029; Awards 2606145, 2606146, and 2606147; total collaborative award: \$749,667)
 - **PIs:** Marcio H. Giacomoni (University of Texas at San Antonio), Ahmad F. Taha (Vanderbilt University), and Jose G. Vasconcelos (Auburn University).
 - **Role:** Proposal contributor and project collaborator. Contributed the hydraulic conceptualization and governing equations for multiphase air–water filling and reviewed the modeling and control methodology.
- **HyREx: Enhancing Hydro-Resilience of Brazilian Catchments to Extreme Events in a Changing World** (FAPESP Process 2025/16429-3, 2026–2031; R\$ 520,368.25 + US\$ 20,700 + two undergraduate, two M.Sc., and one TT-IV fellowships)
 - **PI:** Andre Simoes Ballarin, University of Sao Paulo, Sao Carlos School of Engineering.
 - **Role:** Project collaborator on rainfall temporal structure, climate-change effects on flood extremes, and updated flood-design methods.
- **Funded project: Improving Continuous Streamflow Simulations for Beaver Creek, with Special Focus on Wet Winters** (Salt River Project, 2025; awarded \$129,342)
 - **PI:** Peter A. Troch; **Co-Investigator:** Patrick D. Broxton.
 - **Role:** Project researcher. Developed and calibrated the hsB-SM catchment model coupled with SnowPALM, prepared hydrometeorological and landscape data, and evaluated streamflow simulations with emphasis on wet winters.
- **Proposal submission: Improving Continuous Streamflow Simulations for Beaver Creek, with Special Focus on Wet Winters** (submitted to Salt River Project, 2026; proposed budget: \$80,000; not funded)
 - **PI:** Peter A. Troch; **Co-Investigator:** Patrick D. Broxton.
 - **Role:** Full proposal co-author with Peter A. Troch and Patrick D. Broxton. Co-wrote the technical narrative, preliminary analysis, and work plan for coupling SnowPALM with hsB-SM to improve streamflow simulation during wet winters.
- **Proposal development: Multi-Contaminant Exposure in Moab's Backwaters under Climate Change with Integrated Surface–Groundwater and Transport Modeling** (developed for Keck Foundation consideration, 2025; estimated budget: \$1,010,660)
 - **PI:** Peter A. Troch.
 - **Role:** Proposal co-author and hydrologic–hydrodynamic modeler. Co-developed the monitoring and modeling plan coupling surface flow, river–aquifer exchange, and contaminant transport to estimate exposure to ammonia, uranium, salinity/TDS, and nitrate under historical and climate-change scenarios.
- **Analysis of Unit Hydrograph Models for San Antonio Watersheds** (approximate budget: \$49,000; not funded)
 - **PI** - Dr. Marcio H. Giacomoni
 - **Tasks:** I helped to define the methods used to systematically assess the best unit hydrograph theory used for San Antonio observed data. The idea is that the standard PRF 484 unit hydrograph was oversizing the detention ponds in the city. Therefore, investigating the best-fitted unit hydrograph would reduce the associated costs of low-impact development designs.
- **Curbing Climate Change-Induced Floods via Control Theory for Urban Drainage Systems** (estimated budget: \$250,000; not funded)
 - **PI's** - Dr. Marcio H. Giacomoni and Dr. Ahmad F. Taha
 - **Tasks:** Our previous **research** was the starting point of this proposal. We developed a real-time control model that can potentially increase the efficiency of stormwater systems for flood control. In this proposal, we plan to advance the mathematical modeling to include new stormwater systems (e.g., tunnels, dividers) and actuators (e.g., pumps and gates) to optimally control floods via real-time optimization.

Students Mentored and Research Collaborations

* indicates current or former mentee.

- **Alicia Nguyen*** (Ph.D. student, Stanford University, 2026–pres.): composite hazards in India.
- **Shikshita Bhandari*** (Ph.D. student, Stanford University, 2026–pres.): flood effects on socioeconomics in India.
- **Ricardo Augusto de Farias Freire Filho*** (B.Sc., UFPE, 2025): co-advised undergraduate thesis on hydraulic sizing of stepped channels under skimming-flow conditions; developed and benchmarked an Excel-based design routine against SisCCoH.
- **Maria A. R. A. Castro*** (M.Sc./Ph.D., USP, 2023–pres.): Monte Carlo dam-break ensembles; trade-offs across LI/diffusive/full momentum solvers.
- **Mateo Hernandez*** (M.Sc., USP, 2023–pres.): closed-loop flood control with distributed hydro-hydraulic model; HEC-RAS vs HydroHP comparisons.
- **Jalihal Vijay*** (B.Sc., Vanderbilt, 2023–2024): flood modeling in Bellandur - India.
- **Artur A. T. Brasil** (Ph.D., UTSA, 2022–pres.): permeable pavements field study, digital twin forecasting for LIDs.
- **Matheus dos Santos*** (M.Sc., USP, 2022–2024): MPC in urban reservoirs using RTC-Stormwater; prediction-horizon sensitivity.
- **Milena Rosa*** (B.Sc., USP, 2022–2024): HydroPol2D in data-scarce urban basin; pollutant fate/transport mapping, senior project supervision.
- **Enrico Pavan*** (B.Sc., USP, 2022–2023): large-basin HydroPol2D setup; undergraduate research supervision.
- **Lucas F. Roysen*** (B.Sc., USP, 2022–2023): real-time reservoir control; senior project supervision; RTC-Stormwater applications.
- **Luis M. C. Rápalo*** (Ph.D., USP, 2021–2025): HydroPol2D chapters, urban flood forecasting digital twin, flood inundation mapping, flood damage analysis, and serious-games for risk perception.
- **Ivan Cuervo*** (M.Sc., UTSA, 2020–2022): field monitoring (autosamplers, weirs), water-quality analyses, 2-D runoff modeling.
- **Vivian Jochelavicius*** (B.Sc., USP, 2019): IDF curves under climate scenarios; design implications for compensatory techniques.
- **Hugo de Oliveira Fagundes** (UNICAMP, 2026–pres.): temporal distribution of rainfall extremes in Brazil.
- **Andre Simoes Ballarin** (USP, 2025–pres.): climate change impacts on floods and updated design methods to mitigate climate-change rainfall extremes.
- **Paulo T. S. de Oliveira** (UNC Charlotte, 2023–pres.): hydrodynamic-hydrologic modeling, climate change impacts on floods, and updated design methods to mitigate climate-change rainfall extremes.
- **César A. F. do Lago** (Ph.D., UTSA/USP, 2018–pres.): hydrodynamic modeling, deep learning flood downscaling, ANN WSE prediction; co-developer in HydroPol2D applications.
- **Marina B. de Macedo** (Ph.D., UNIFEI/USP, 2018–2022): LIDs, climate/urbanization impacts, modular bioretention design; multiple joint papers.

Professional Service

Ad hoc reviewer work	Performed 73 verified peer reviews for 27 manuscripts and grant submissions across journals including <i>Journal of Hydrology</i> , <i>Water Resources Research</i> , <i>Environmental Modelling & Software</i> , <i>International Journal of Disaster Risk Reduction</i> , <i>Water Research</i> , <i>Journal of Flood Risk Management</i> , <i>Hydrology and Earth System Sciences</i> , and <i>Journal of Hydrologic Engineering</i> .
Conference panelist reviewer	Panelist reviewer for El Día del Agua y la Atmósfera , University of Arizona, March 25, 2025.
Memberships	American Geophysical Union (AGU), 2021–2022 and 2024–present; American Society of Civil Engineers (ASCE), 2020–2021; Brazilian Water Resources Association (ABRH), 2019–2020 and 2021–2022.

Public Outreach

Engineering education	Founder of Engenheiro Planilheiro , a Portuguese-language civil and environmental engineering website with more than 60 numerical tools, 30+ technical articles, and more than 90,000 downloads from users in more than 10 countries.
YouTube	Created Portuguese-language educational videos on civil and environmental engineering topics through Engenheiro Planilheiro .

Press

Hydrologic models	<i>Modelos hidrológicos desenvolvidos por alunos da USP ajudam a prever riscos de desastres</i> [English: Hydrologic models developed by USP students help predict disaster risks], Jornal da USP , Aug. 28, 2024.
Public media	<i>Alunos da USP criam ferramenta para prever desastres naturais</i> [English: USP students create a tool to predict natural disasters], Olhar Digital , Aug. 30, 2024; <i>Alunos da USP criam ferramenta que prevê desastres</i> [English: USP students create a tool that predicts disasters], TudoCelular , 2024.
EESC-USP coverage	<i>Novos modelos comunitários desenvolvidos por alunos ajudam com maior previsibilidade de riscos de desastres</i> [English: New community models developed by students improve disaster-risk predictability], EESC-USP, Aug. 6, 2024.
Thesis award	Coverage of the 13th <i>Prêmio Tese Destaque USP</i> [USP Distinguished Thesis Award], 2024: • Jornal da USP • EESC-USP • EESC-USP result announcement • EESC-USP PPG • PRPG-USP

References

- **Steven Gorelick**, Professor, Department of Earth System Science, Stanford University.
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- **Peter A. Troch**, Professor, Department of Hydrology and Atmospheric Sciences, University of Arizona.
Email: patroch@arizona.edu. [Google Scholar](#), [Website](#).
- **Paul D. Bates**, Professor, School of Geographical Sciences, University of Bristol.
Email: paul.bates@bristol.ac.uk. [Google Scholar](#), [Website](#).
- **Paulo T. S. de Oliveira**, Associate Professor, Civil & Environmental Engineering, University of North Carolina at Charlotte.
Email: polivei2@charlotte.edu. [Google Scholar](#), [Website](#).
- **Ahmad F. Taha**, Associate Professor, Civil and Environmental Engineering and Electrical and Computer Engineering, Vanderbilt University.
Email: ahmad.taha@vanderbilt.edu. [Google Scholar](#), [Website](#).
- **Marcio H. Giacomoni**, Professor, School of Civil & Environmental Engineering, and Construction Management, The University of Texas at San Antonio.
Email: marcio.giacomoni@utsa.edu. [Google Scholar](#), [Website](#).
- **Eduardo Mario Mendonido**, Professor, Department of Hydraulics and Sanitation, Sao Carlos School of Engineering, University of Sao Paulo.
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